



SIH 2025 · PS #26015 · SOFTWARE EDITION

WatershedVision

Converting field photographs and satellite data into actionable watershed decisions.

Problem Statement ID:	26015
Problem Statement Title:	Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes
Ministry / Department:	Department of Land Resources (DoLR), Ministry of Rural Development
Theme:	Agriculture, FoodTech & Rural Development
PS Category:	Software
Team Name / ID:	WatershedVision Team [SIH-2025-WV01]
Institution:	[Registered Institution / College Name]
Team Members & Depts:	6 Members (CSE, AI & Data Science, Civil/GIS, IT)
Mentor:	[Faculty / Industry Domain Mentor Name]



Sentinel-2 L2A Ingested SRTM 30m DEM Anti-Spoofing Multimodal Civil AI

IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)



⚠️ PROBLEM STATEMENT REALITIES

- **Massive Photo Archives:** Thousands of geo-tagged ground images collected across rural districts without automated categorization.
- **Slow Manual Auditing:** Field verification teams take weeks to identify damaged or silted structures.
- **Data Disconnection:** Field photos, satellite rasters, and cadastral boundaries exist in disconnected silos.
- **Reactive Maintenance:** Officers lack real-time decision support to triage repair budgets before monsoons.

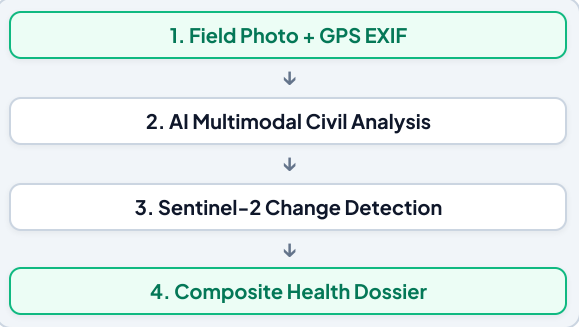
💡 PROPOSED SOLUTION: WATERSHEDVISION COMMAND CENTRE

- **Intelligent Command Portal:** Merges field photographs, GPS EXIF metadata, satellite imagery, and multimodal AI.
- **Civil Structure Classification:** Identifies check dams, farm ponds, and bunds with structural integrity and siltation % scoring.
- **Satellite Outcome Tracking:** Computes multi-temporal NDVI and NDWI deltas over multi-year treatment periods.
- **Evidence-Based Reporting:** Generates statutory MoRD/DoLR audit dossiers with cryptographic verification seals.

🚀 INNOVATION & TECHNICAL UNIQUENESS

- **Passive Archive to Active Intelligence:** Automatically classifies and grades legacy photo archives without manual tagging.
- **3D Terrain Anti-Spoofing:** Cross-checks smartphone EXIF altitude against NASA SRTM 30m DEM to eliminate mock GPS tampering.
- **Cadastral Evidence Linking:** Enforces PostGIS point-in-polygon checks against registered micro-watershed boundaries.
- **Risk-Based Inspection Triage:** Ranks physical interventions by siltation severity and capacity retention.

DECISION WORKFLOW FLOWCHART:



TECHNICAL APPROACH

Technologies to be used & Implementation Methodology



EDITABLE SYSTEM ARCHITECTURE & DATA FLOW PIPELINE



TECHNOLOGIES & PRODUCTION STACK

- **Frontend Presentation:** React 18, Vite, Tailwind CSS, Leaflet JS, Recharts data visualization.
- **Backend API Services:** Python 3.11, FastAPI (async/await), Uvicorn ASGI, Docker containerization.
- **Geospatial Database:** PostgreSQL 15 + PostGIS spatial extensions for polygon boundaries and OGC queries.
- **Earth Observation Processing:** Google Earth Engine (GEE), Copernicus Sentinel-2 MSI (10m), Landsat 8/9.
- **Multimodal Vision AI:** Multimodal Gemini Vision API + PyTorch feature classifiers for civil structure inspection.
- **Data & Protocols:** RESTful JSON APIs, OGC GeoJSON FeatureCollections, GDAL / Rasterio geoprocessing.

IMPLEMENTATION METHODOLOGY & LIVE PROTOTYPE

- **Automated EXIF Extraction:** Extracts lat/long, altitude, and timestamps from uploaded JPEG/HEIC/PNG survey photos.
- **3D GPS Anti-Spoofing:** Cross-examines photo elevation with SRTM 30m DEM; flags discrepancies >35m as tamper alerts.
- **Satellite Change Detection:** Generates cloud-masked pre-monsoon vs. post-monsoon NDVI/NDWI raster layers.
- **Bilingual Field Accessibility:** Full English and Hindi (हिंदी) localization for Gram Panchayat and SLNA officers.

Live Satellite Analytical Products:
10m NDVI Vegetation Delta · NDWI Water Surface · Strahler Stream Orders · KDE Intervention Heatmaps



FEASIBILITY AND VIABILITY

Feasibility Analysis, Potential Challenges & Mitigation Strategies



✓ FEASIBILITY ANALYSIS

- **Proven Open-Source Stack:** Relies on battle-tested frameworks (FastAPI, PostGIS, Leaflet) eliminating proprietary license lock-in.
- **Deployable Web Platform:** Live working prototype deployed on cloud container infrastructure with zero client-side installs.
- **High-Throughput Spatial Engine:** PostGIS handles thousands of micro-watersheds with sub-second spatial boundary lookups.
- **Cloud Satellite Computation:** Google Earth Engine offloads heavy multispectral raster math from local field machines.
- **Asynchronous Worker Pipeline:** Non-blocking AI analysis queues ensure high UI responsiveness during heavy upload batches.

⚠ POTENTIAL CHALLENGES & RISKS

- **Missing or Stripped GPS:** Certain messaging apps and older smartphone cameras strip EXIF geolocation metadata.
- **Varying Photo Quality:** Blurry, low-light, or occluded photographs taken by field surveyors during stormy conditions.
- **Monsoon Cloud Cover:** Optical Sentinel-2 imagery obscured by heavy monsoon cloud cover during critical post-rain cycles.
- **External API Failures:** Intermittent network outages or rate limitations on commercial cloud AI endpoints.
- **Domain Interpretation Risk:** Risk of false positives in automated civil structure siltation predictions.

🛡 MITIGATION STRATEGIES

- **Interactive Geotag Correction:** Surveyors can manually verify or drop coordinates on cadastral boundary maps with audit logs.
- **Confidence Gating:** Predictions under 75% confidence are flagged for human inspection by the district engineer.
- **Temporal Median Compositing:** Multi-date cloud-masked compositing (May-Oct) penetrates cloud gaps using QA bands.
- **Graceful Fallback & Caching:** Local spatial caching with clearly labelled offline prototype data structures.
- **Human-in-the-Loop Governance:** AI provides decision support; financial sanctions remain with designated officers.

SCALE & ROADMAP: Micro-Watershed → District (DWDC) → State Nodal Agency (SLNA) | Offline-First PWA Sync | Bhuvan & SRISHTI Integration

Multi-Hazard Ready: Erosion · Flood · Drought

IMPACT AND BENEFITS

Stakeholder Value, Operational Outcomes & Environmental Sustainability



TARGET BENEFICIARIES

- **District Watershed Officers:** Holistic dashboard tracking intervention progress and fund allocation.
- **Field Surveyors & Supervisors:** Rapid smartphone image upload with instant geofence and elevation validation.
- **GIS & Remote Sensing Cells:** Ready-to-use vector drainage, NDVI deltas, and intervention density heatmaps.
- **State & Central Administrators:** Standardized, audit-ready compliance reporting across thousands of projects.
- **Village Gram Panchayats:** Bilingual Hindi/English views fostering grassroots community participation.

OPERATIONAL & GOVERNANCE BENEFITS

- **Faster Photo Auditing:** Replaces manual visual scanning with automated classification and structural grading.
- **Anti-Fraud Geotag Verification:** Point-in-polygon and DEM elevation checks eliminate fraudulent location reporting.
- **Objective Prioritization:** Ranks physical interventions by siltation severity and capacity retention.
- **Transparent Statutory Reporting:** One-click generation of official DoLR/MoRD format dossiers with QR verification.
- **Data Fusion:** Bridges the gap between macro satellite trends and micro ground-level civil conditions.

ENVIRONMENTAL & ECOLOGICAL OUTCOMES

- **Plantation Survival Tracking:** Measures canopy cover recovery over 3-year periods to verify afforestation success.
- **Soil Erosion & Gully Monitoring:** Identifies drainage channels requiring emergency gully plugs or contour bunds.
- **Surface Water Storage Expansion:** Monitors water body expansion and desiltation outcomes across post-monsoon cycles.
- **Climate Resilience:** Strengthens groundwater recharge and agricultural drought resilience for smallholder farmers.
- **Sustainable Basin Planning:** Supports carrying-capacity analysis using Strahler stream order modeling.

“WatershedVision helps administrators move from reactive inspection to proactive, data-driven watershed management.”

RESEARCH AND REFERENCES

Details / Links of Reference & Research Work, Project Verification



OFFICIAL STANDARDS, SCIENTIFIC LITERATURE & DOCUMENTATION

1. WDC-PMKSY 2.0 Operational Guidelines
Department of Land Resources (DoLR), Ministry of Rural Development, Government of India

2. Bhuvan SRISHTI & DRISHTI Geospatial Monitoring Framework
National Remote Sensing Centre (NRSC) / Indian Space Research Organisation (ISRO)

3. Copernicus Sentinel-2 MSI Multi-Spectral Technical Documentation
European Space Agency (ESA) · 10m Multi-Band Optical Surface Reflectance (L2A)

4. USGS Landsat 8-9 Surface Reflectance Tier 1 Specification
U.S. Geological Survey & NASA · Calibrated Earth Observation Spectral Archive

5. Google Earth Engine: Planetary-Scale Geospatial Analysis
Gorelick, N. et al. (Remote Sensing of Environment, 2017) · Cloud Geospatial Processing

6. PostGIS 3.4 Spatial Database & OGC Simple Features Standard
Open Geospatial Consortium (OGC) · Standardized Point-in-Polygon & Spatial Query Engine

INTERACTIVE LIVE VERIFICATION



Scan to Launch Live Web Application
<https://watershed-vision.vercel.app/>
<https://github.com/psryogeshwar-14/watershed-vision>

[FastAPI](#) [React](#) [PostGIS](#) [Sentinel-2](#) [Bilingual](#)

GOVERNANCE NOTE: "AI predictions support screening and prioritisation. Final engineering, funding, and maintenance decisions remain with authorised officers."